

ANALYSTLAB AFRICA EXPERIENCE LAB

Official Week 4 Project Summary — Machine Learning Track

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Track: Machine Learning / Data Science / ML Engineering

Project Title: HealthConnect Clinic Appointment Attendance Optimization

Date: August 2026

1. Specific Problem Addressed

HealthConnect Clinic experiences significant operational disruptions and unrecoverable idle capacity caused by patient appointment no-shows (48.46% baseline non-attendance rate across 5,000 appointment records). The Machine Learning track addresses this problem by developing an automated prediction and decision engine to identify high-risk non-attendance prior to scheduled appointment times, enabling proactive multi-channel interventions and schedule optimization.

2. Resources Utilized

1. **HealthConnect_Appointment_Data.csv:** 5,000 appointment records across 1,696 unique patients, containing demographic, scheduling, historical attendance, and reminder metrics.
2. **HealthConnect_Data_Dictionary.xlsx:** Complete field definition dictionary specifying data types, role constraints, and descriptions.
3. **HealthConnect_Clinic_Knowledge_Base.docx:** Clinic operational rules, cancellation policies, and administrative procedures.

3. Key Observations from Week 4

- **Target Distribution:** No-Show (48.46%, $N = 2,423$), Attended (46.28%, $N = 2,314$), Cancelled (5.26%, $N = 263$).
- **Booking Lead Time Effect:** Appointments booked ≥ 30 days in advance exhibit over 65% no-show rates, compared to $< 25\%$ for appointments booked within 3 days.
- **Historical Recidivism:** Patients with ≥ 2 previous no-shows exhibit a $> 75\%$ probability of missing subsequent appointments.
- **Data Quality:** `reminder_channel` missingness (27.32%) is NMAR (corresponds to no reminder sent); `waiting_time_minutes` presents a critical target leakage risk and must be excluded.

4. Proposed Technical & System Approach

We propose a supervised binary classification model $P(\text{No-Show} \mid X)$ integrated into a 5-stage production architecture: Data Ingestion $\rightarrow$ Validation & Feature Store $\rightarrow$ Inference Microservice $\rightarrow$ Actionable Decision Engine $\rightarrow$ MLOps Logging. Patients are stratified into High, Moderate, and Low risk tiers to trigger automated WhatsApp/SMS confirmation workflows.

5. Key Considerations, Risks & Constraints

- **Data Leakage Boundary:** Explicit exclusion of post-arrival variables (`waiting_time_minutes`).
- **Patient Clustering:** GroupKFold cross-validation grouped by `patient_id` to prevent temporal behavior leakage across multiple patient appointments.
- **Ethical Compliance:** Ensuring equitable prediction accuracy across age and gender subgroups without service discrimination.

6. Proposed Focus for Week 5

In Week 5, work will progress to **Data Preprocessing, Feature Engineering & Baseline Model Prototyping**:

1. Constructing automated scikit-learn preprocessing and imputation pipelines.
2. Engineering behavioral features (`historical_noshow_ratio`, lead time interactions).
3. Benchmarking baseline classification models (Logistic Regression, Decision Trees, Random Forest, XGBoost, LightGBM).
4. Tuning classification decision thresholds for high recall performance.